TERMS OF REFERENCE

Business Intelligence Consulting Assignment

OW Yellow Caps

Executive Insights from NYC Yellow Taxi Trips

January-March 2025

Document version	1.0
Client	OW Yellow Caps
Engagement type	Business Intelligence and Executive Reporting
Submission audience	Executive and non-technical stakeholders
Primary data source	NYC TLC Yellow Taxi Trip Records
Supporting metadata	TLC Taxi Zone lookup and Yellow Taxi data dictionary

Prepared for student consulting groups

1. OW Yellow Caps (the Client)

OW Yellow Caps is a fictional urban mobility stakeholder commissioning this assignment to obtain clear, executive-level insights from New York City Yellow Taxi operations during January-March 2025. Each student group will act as an external Business Intelligence company and is expected to convert a large operational dataset into trustworthy, concise, and decision-oriented reporting for managers who are not data specialists. The source data are the official NYC Taxi and Limousine Commission (TLC) Yellow Taxi trip records and the associated taxi zone metadata [1-3].

2. Introduction and Project Background

The project is designed to simulate a real consulting engagement. The client requires descriptive and diagnostic intelligence, not predictive modeling, machine learning, or hypothesis testing. Students are expected to inspect the supplied trip records, evaluate their quality, build an analysis-ready dataset, and communicate the data's operational story through well-designed visualizations and a concise narrative. TLC describes these records as trip submissions from authorized technology providers and provides official field definitions and taxi zone metadata to support interpretation [1-3].

The main business question is: what do the January-March 2025 Yellow Taxi records reveal about demand, geography, trip characteristics, and fare behavior, and what data quality issues materially affect confidence in those insights?

3. Project Objective

Each group shall deliver a Business Intelligence package that helps executive stakeholders understand what is happening, where it is happening, when it is happening, and which operational patterns deserve attention. The work must rely on exploratory data analysis (EDA) and visualization. The final outputs should be suitable for executive review, operational planning discussions, and non-technical briefings.

4. Scope of Work

- Inspect the supplied data and supporting metadata.
- Validate fields, identify irregularities, and document data quality risks.
- Preprocess the data and create a cleaned analytical layer or curated summary tables.
- Produce executive-facing insights using univariate, bivariate, and multivariate EDA.
- Develop business-friendly visualizations and a dashboard or reporting product.
- Explain the main limitations, assumptions, and sensitivity checks.

5. Expected Deliverables

5.1 Technical Preprocessing Report

A separate technical report covering schema review, validation rules, missingness, duplicates, cleaning decisions, sensitivity checks, and limitations. Target length: 10 pages max.

5.2 Executive Insights Report

A concise business report for non-technical readers containing context, KPIs, major insights, supporting charts, business implications, and a short summary of the main data issues that matter for interpretation. Target length: 10 pages max.

5.3 Executive Presentation Deck

A senior-management presentation summarizing scope, findings, caveats, and recommendations. Target length: 20 slides max and 15 minutes presentation.

5.5 Reproducible Code Package

A clean submission folder containing scripts, dependencies, and a short README explaining how the outputs were produced.

5.6 Analytical Output Data

A cleaned analytical dataset or a documented set of curated summary tables used in the final BI product.

6. Required Analytical Coverage

6.1 Univariate EDA

At minimum, teams should assess trip volume over time; the distributions of trip distance, trip duration, fare amount, total amount, and passenger count; and the frequencies of payment type and rate code.

6.2 Bivariate EDA

At a minimum, teams should examine hour of day versus trip volume, geography versus revenue or trip count, payment type versus tip or total amount, distance versus total amount, and weekday versus demand.

6.3 Multivariate EDA

At minimum, teams should investigate combinations such as time x geography x demand, geography x payment type x revenue, distance x duration x total amount, month x hour x borough or zone, and pre/post January 5, 2025 patterns where cbd_congestion_fee is relevant [2].

7. Milestones and Suggested Timeline

Week	Milestone
1-2	Data familiarization, metadata review, and project framing
3-4	Data validation and preprocessing plan
5	Submission of the preprocessing progress memo
6-8	Core EDA and visualization development
9	Interim client-style presentation
10-11	Refinement of report and dashboard
12	Final submission and executive presentation

8. Submission Format and Reporting Standards

All written submissions must use A4, 12-point readable font, 1.5-line spacing, 1-inch margins, numbered sections, page numbers, figure titles, labeled axes, and a professional business tone. The Executive Insights Report and deck should be designed for readers who may only spend a few minutes on each page or slide. Dense technical discussion belongs in the preprocessing report or appendix.

9. Evaluation and Acceptance Criteria

Criterion	Weight
Data preprocessing rigor and transparency	25%
Quality of EDA and insight generation	25%
Relevance to business and executive audience	20%
Quality and clarity of visualizations	15%
Reproducibility and organization of deliverables	10%
Professionalism of reporting and presentation	5%

A strong submission will be analytically sound, visually clear, honest about limitations, business-relevant, reproducible, and easy for non-technical managers to understand.

Appendix A. Guidelines and Rules

Students must not assume that all source records are valid. At minimum, each team is expected to examine field validity, missingness, temporal consistency, cross-field logical consistency, fare and payment consistency, location validity using taxi zones, duplicates and near-duplicates, and the sensitivity of conclusions to preprocessing choices [1]-[3].

The official Yellow Taxi data dictionary states that tpep_pickup_datetime is when the meter was engaged, tpep_dropoff_datetime is when the meter was disengaged, trip_distance is the elapsed distance in miles reported by the taximeter, RatecodeID valid values are 1, 2, 3, 4, 5, 6, and 99, store_and_fwd_flag should be Y or N, payment_type valid values are 0, 1, 2, 3, 4, 5, and 6, tip_amount is automatically populated for credit-card tips, total_amount does not include cash tips, airport_fee applies only to pickup at LaGuardia and John F. Kennedy Airports, and cbd_congestion_fee begins on January 5, 2025 [2].

Each group must include at least one short sensitivity analysis showing whether a reasonable alternative cleaning decision materially changes the headline business conclusions.

The executive report and presentation must include a brief summary of the main data quality issues that affect interpretation, while detailed remediation steps belong in a separate technical preprocessing report.

10. Data Dictionary (Core Fields for This Assignment)

The authoritative field definitions are maintained by the TLC [2]. For this assignment, groups are expected to understand and use the following core fields.

Field Name	Description
VendorID	A code indicating the TPEP provider that provided the record. 1 = Creative Mobile Technologies, LLC 2 = Curb Mobility, LLC 6 = Myle Technologies Inc 7 = Helix
tpep_pickup_datetime	The date and time when the meter was engaged.
tpep_dropoff_datetime	The date and time when the meter was disengaged.
passenger_count	The number of passengers in the vehicle.
trip_distance	The elapsed trip distance in miles reported by the taximeter.
RatecodeID	The final rate code in effect at the end of the trip. 1 = Standard rate 2 = JFK 3 = Newark 4 = Nassau or Westchester 5 = Negotiated fare 6 = Group ride 99 = Null/unknown
store_and_fwd_flag	This flag indicates whether the trip record was held in vehicle memory before sending to the vendor, aka "store and forward," because the vehicle did not have a connection to the server. Y = store and forward trip N = not a store and forward trip
PULocationID	TLC Taxi Zone in which the taximeter was engaged.
DOLocationID	TLC Taxi Zone in which the taximeter was disengaged.
payment_type	A numeric code signifying how the passenger paid for the trip. 0 = Flex Fare trip 1 = Credit card 2 = Cash 3 = No charge 4 = Dispute 5 = Unknown 6 = Voided trip
fare_amount	The time-and-distance fare calculated by the meter. For additional information on the following columns, see https://www.nyc.gov/site/tlc/passengers/taxi-fare.page
extra	Miscellaneous extras and surcharges.
mta_tax	Tax that is automatically triggered based on the metered rate in use.
tip_amount	Tip amount – This field is automatically populated for credit card tips. Cash tips are not included.
tolls_amount	Total amount of all tolls paid in trip.
improvement_surcharge	Improvement surcharge assessed trips at the flag drop. The improvement surcharge began being levied in 2015.
total_amount	The total amount charged to passengers. Does not include cash tips.
congestion_surcharge	Total amount collected in trip for NYS congestion surcharge.
airport_fee	For pick up only at LaGuardia and John F. Kennedy Airports.
cbd_congestion_fee	Per-trip charge for MTA's Congestion Relief Zone starting Jan. 5, 2025.

Important note: The trip_id column was added as a proxy for row numbers. It helps in identifying the row number when documenting issues in the dataset. This column is not original in the dataset.

11. References

[1] New York City Taxi and Limousine Commission. TLC Trip Record Data. Available at: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

[2] New York City Taxi and Limousine Commission. Data Dictionary - Yellow Taxi Trip Records. March 18, 2025. Available at: https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf

[3] U.S. Data Catalog / NYC TLC. NYC Taxi Zones. Available at: https://catalog.data.gov/dataset/nyc-taxi-zones-131e4